EUSIPCO 2026 · POSTER SESSION

NTNU

Cost-aware graph augmentation for robust networks via effective resistance

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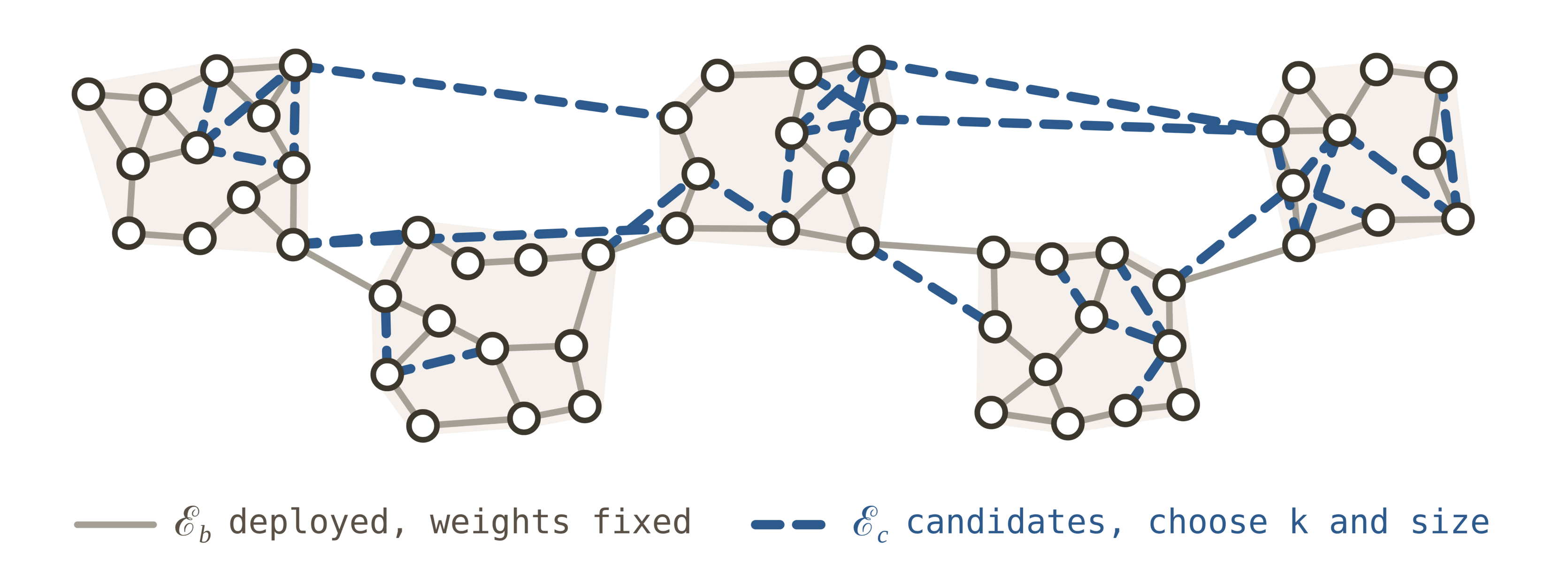
01 Optimizing a pre-deployed network

Total effective resistance R_{tot} (the Kirchhoff index) is a tractable spectral proxy for robustness. Reducing it on a *deployed* weighted network requires deciding **which links to add and how strongly to weight them**. This couples a discrete selection problem with a continuous sizing problem under a single global budget. Costs vary across candidate links, combining a fixed installation overhead with a length-dependent penalty.



The setup. We select exactly k candidates and size each within $[w_{\min}, w_{\max}]$. Every candidate carries its own unit price c_i and the total spend cannot exceed C.

OBJECTIVE

$$\mathbf{L}(\mathbf{w}) = \mathbf{L}_b + \underbrace{\sum_{e\in\mathcal{E}_c} w_e\,\mathbf{a}_e\mathbf{a}_e^\top}_{\mathbf{L}_a(\mathbf{w})}, \qquad \mathbf{M}(\mathbf{w}) \coloneqq \mathbf{L}(\mathbf{w}) + \frac{\mathbf{1}\mathbf{1}^\top}{n} \succ 0$$

Here $\mathbf{L}_b$ is the base Laplacian and $\mathbf{L}_a(\mathbf{w})$ the augmentation Laplacian, where candidate e enters through its incidence vector $\mathbf{a}_e$.

$$R_{\mathrm{tot}}(\mathbf{w}) = n\,\mathrm{Tr}\left[\mathbf{M}^{-1}(\mathbf{w})\right] - n$$

02 Joint selection and sizing

MIXED-INTEGER PROGRAM $(\mathcal{P}_0)$

$$\begin{array}{lll} \underset{\mathbf{w},\,\mathbf{z}}{\text{minimize}} & R_{\mathrm{tot}}(\mathbf{w}) & \text{robustness of the upgraded network} \\ \text{subject to} & \mathbf{c}^\top\mathbf{w} \le C & \text{one global budget} \\ & z_i w_{\min} \le w_i \le z_i w_{\max} & \text{perspective box, } z_i = 0 \Rightarrow w_i = 0 \\ & \mathbf{1}^\top\mathbf{z} = k, \quad z_i \in \{0,1\} & \text{exactly } k \text{ new links} \end{array}$$

03 SDP relaxation and feasible recovery

The binary variables z_i render $(\mathcal{P}_0)$ NP-hard, and evaluating the inverse in R_{tot} requires an $\mathcal{O}(n^3)$ matrix factorization. Relaxing $\mathbf{z}$ to $[0,1]^p$ removes the combinatorial difficulty. Since $\mathbf{M}(\mathbf{w}) \succ 0$ for a connected base graph, $\mathrm{Tr}(\mathbf{M}^{-1}(\mathbf{w})) \le \mathrm{Tr}(\mathbf{R})$ can be imposed through a Schur complement, eliminating the explicit inverse and yielding a convex formulation.

SEMIDEFINITE SURROGATE $(\mathcal{P}_1)$

$$\begin{array}{ll} \underset{\mathbf{R},\,\mathbf{w},\,\mathbf{z}}{\text{minimize}} & n\,\mathrm{Tr}(\mathbf{R}) - n \\ \text{subject to} & \begin{bmatrix} \mathbf{M}(\mathbf{w}) & \mathbf{I} \\ \mathbf{I} & \mathbf{R} \end{bmatrix} \succeq 0 \\ & \mathbf{c}^\top\mathbf{w} \le C \\ & z_i w_{\min} \le w_i \le z_i w_{\max}, \quad i = 1,\dots,p \\ & \mathbf{1}^\top\mathbf{z} = k, \quad 0 \le z_i \le 1, \quad \mathbf{R} \in \mathbb{S}_+^n \end{array}$$

Rounding and repair. Keep the k largest entries of $\mathbf{z}^\star$ and break ties toward cheaper links. Project the weights onto the induced box. If the design is over budget, scale toward $w_{\min}\hat{\mathbf{z}}$ until $\mathbf{c}^\top\hat{\mathbf{w}} \le C$. The recovered design is feasible only when the selected support can carry the minimum weights.

Proposition 1. *Assume the base graph is connected, so* $\mathbf{M}(\mathbf{w}) \succ 0$ *for all feasible* $\mathbf{w}$. *Then* $(\mathcal{P}_1)$ *satisfies* $R_{\mathrm{tot}}^{\mathrm{SDP}} \le R_{\mathrm{tot}}^{\mathrm{MI}}$. *Moreover, at any optimum* $(\mathbf{R}^\star, \mathbf{w}^\star, \mathbf{z}^\star)$ *of* $(\mathcal{P}_1)$, $\mathbf{R}^\star = \mathbf{M}^{-1}(\mathbf{w}^\star)$. *Consequently, the exact optimum of* $(\mathcal{P}_1)$ *is a lower bound on* $(\mathcal{P}_0)$.

Remark 1. *Given a feasible rounded pair* $(\hat{\mathbf{w}}, \hat{\mathbf{z}})$, *we have*

$$R_{\mathrm{tot}}^{\mathrm{SDP}} = \mathrm{val}(\mathcal{P}_1) \le \mathrm{val}(\mathcal{P}_0) = R_{\mathrm{tot}}^{\mathrm{MI}} \le R_{\mathrm{tot}}(\hat{\mathbf{w}}) =: R_{\mathrm{tot}}^{\mathrm{RND}}$$

so the relative ratio $\left(R_{\mathrm{tot}}^{\mathrm{RND}} - R_{\mathrm{tot}}^{\mathrm{SDP}}\right)/R_{\mathrm{tot}}^{\mathrm{SDP}}$ *is an a posteriori gap certificate for the recovered design.*

04 Solved as a cone program

Scaled vectorization stacks the unknowns as $\mathbf{x} = \left[\mathrm{svec}(\mathbf{R})^\top\ \mathbf{w}^\top\ \mathbf{z}^\top\right]^\top$. The constraints take the standard conic form $\mathbf{F}\mathbf{x} + \mathbf{s} = \mathbf{b}$, with $\mathbf{s} \in \mathcal{K}$.

$$\mathcal{K} = \underbrace{\mathbb{S}_+^{2n}}_{\text{LMI}} \times \underbrace{\mathbb{R}_+}_{\text{budget}} \times \underbrace{\mathbb{R}_+^p \times \mathbb{R}_+^p \times \mathbb{R}_+^p \times \mathbb{R}_+^p}_{\text{perspective box and } z \text{ bounds}} \times \underbrace{\{0\}}_{\text{cardinality}}$$

- Solved by ADMM on the homogeneous self-dual embedding, so the reported bound holds to the solver tolerance.
- A cached $\mathbf{LDL}^\top$ **factorization** reduces each affine step to two triangular solves.

05 Up-to-k greedy with rank-one updates

A cost-aware greedy algorithm adds up to k links, stopping early if no remaining candidate is affordable. The Sherman–Morrison rank-one update provides each marginal gain in closed form. The inverse $\mathbf{M}^{-1}$ is formed once and then updated after every insertion rather than recomputed. Each trial weight is priced by a **tentative budget allocation** $s = C_{\mathrm{rem}}/k_{\mathrm{rem}}$.

MARGINAL GAIN, EXACT

$$\Delta R_{\mathrm{tot}} = \frac{n\,w_{(v_i,v_j)}\,b^2(v_i,v_j)}{1 + w_{(v_i,v_j)}\,r(v_i,v_j)}$$

TRIAL WEIGHT

$$w_e \coloneqq \max\{w_{\min},\ \min\{w_{\max},\ s/c_e\}\}$$

from the tentative allocation $s = C_{\mathrm{rem}}/k_{\mathrm{rem}}$

$r(v_i,v_j) \coloneqq \mathbf{a}^\top\mathbf{M}^{-1}\mathbf{a}$ $\qquad$ $b^2(v_i,v_j) \coloneqq \|\mathbf{M}^{-1}\mathbf{a}\|_2^2$ $\qquad$ total cost: $\mathcal{O}\left(n^3 + k(pn + n^2)\right)$

06 Instances and candidate sets

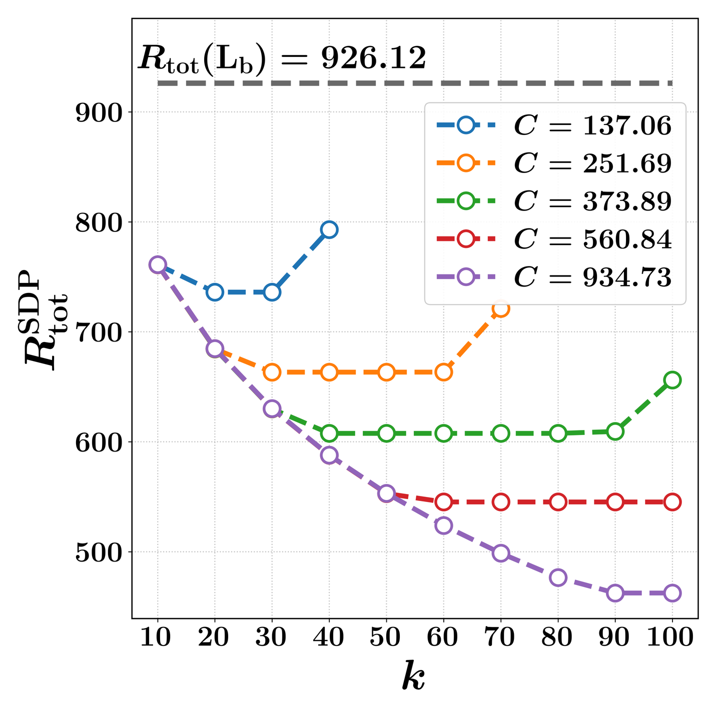
Synthetic. Base graph of $n = 100$ planar vertices with lognormal weights. Candidate weights span the 0.75 to 0.95 quantiles. Three k-means regions give $k_{\mathrm{in}} = 3$ neighbour links and a sparse backbone.

Real-world. California road network with every base edge weight set to one. We sample $n = 500$ nodes and keep the largest connected component. Unused node pairs are candidates in $[1, 5]$.

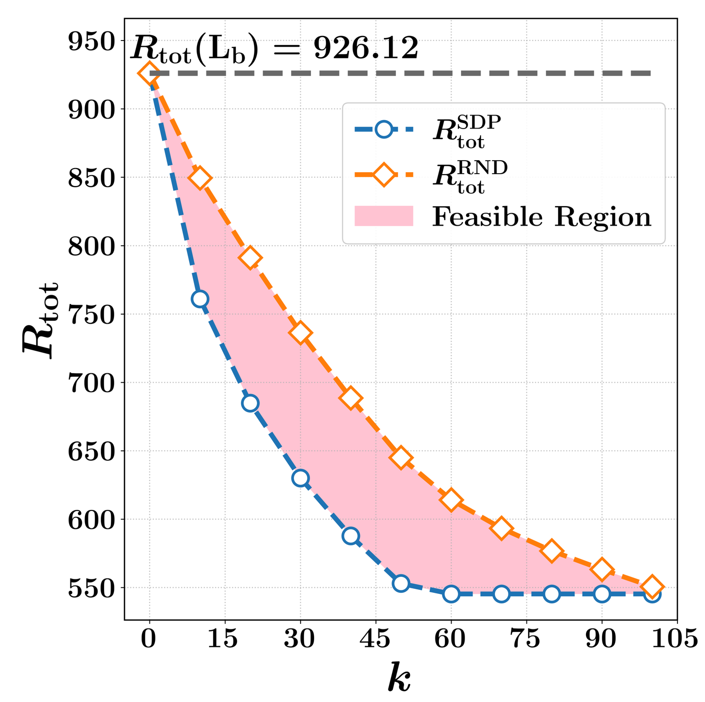
Cost model. Candidate edges follow $\mathbf{c} = c_{\mathrm{fixed}}\mathbf{1} + \beta\mathbf{d}_c$, so each candidate has a per-unit price. Here, c_{fixed} is the baseline per-unit cost, and the candidate link lengths $\mathbf{d}_c$ are normalized.

Budget regimes. Budgets are deduced from the candidate cost structure. The least budget admitting k links is $C_{\min}(k) = w_{\min}\sum_{i=1}^{k} c_{(i)}$, summed over the cheapest candidates. Five levels centre on the anchor $C_{\min}(k_{\max}) = 373.89$, two at the thresholds $k_{\max}/3$ and $2k_{\max}/3$ and two at $1.5\times$ and $2.5\times$.

07 What the budget buys



1a. $R_{\mathrm{tot}}^{\mathrm{SDP}}$ versus k at different budgets.



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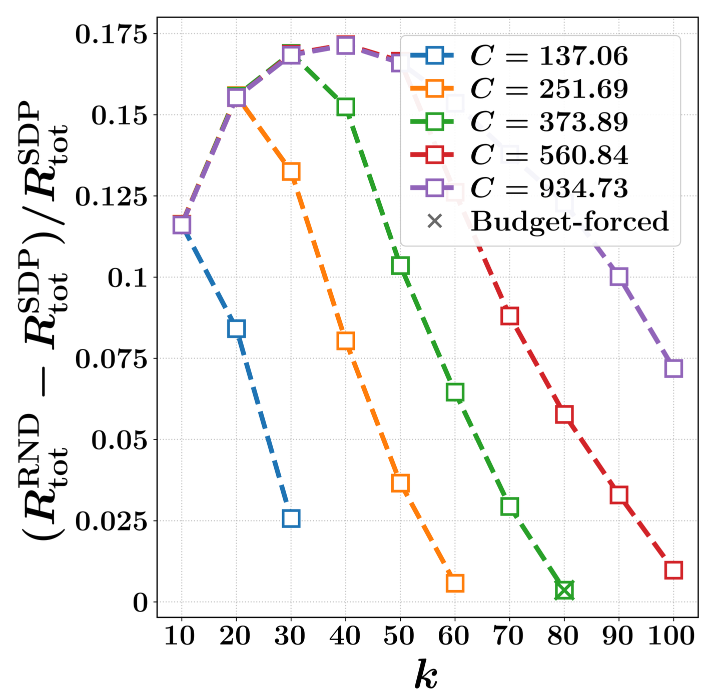
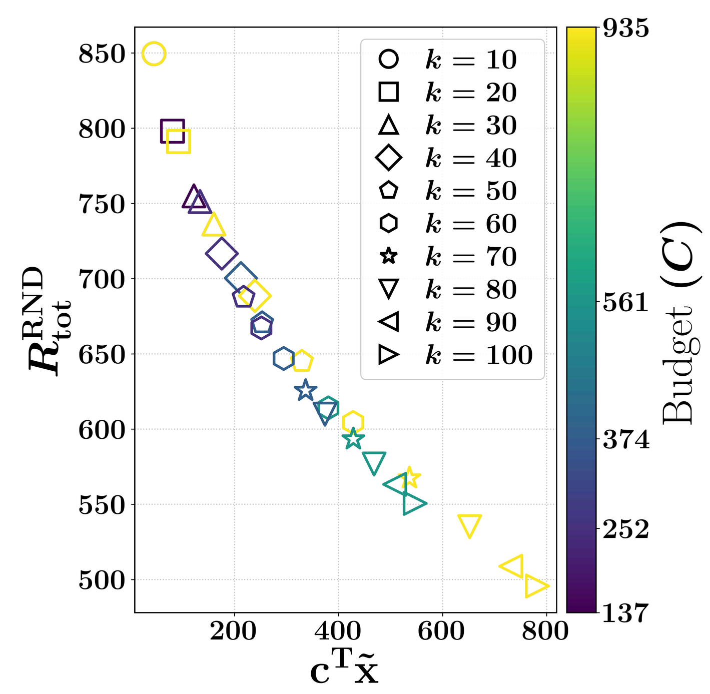


1c. Relative gap $\left(R_{\mathrm{tot}}^{\mathrm{RND}} - R_{\mathrm{tot}}^{\mathrm{SDP}}\right)/R_{\mathrm{tot}}^{\mathrm{SDP}}$ at each budget.



1d. Pareto front against budget spent at fixed k.

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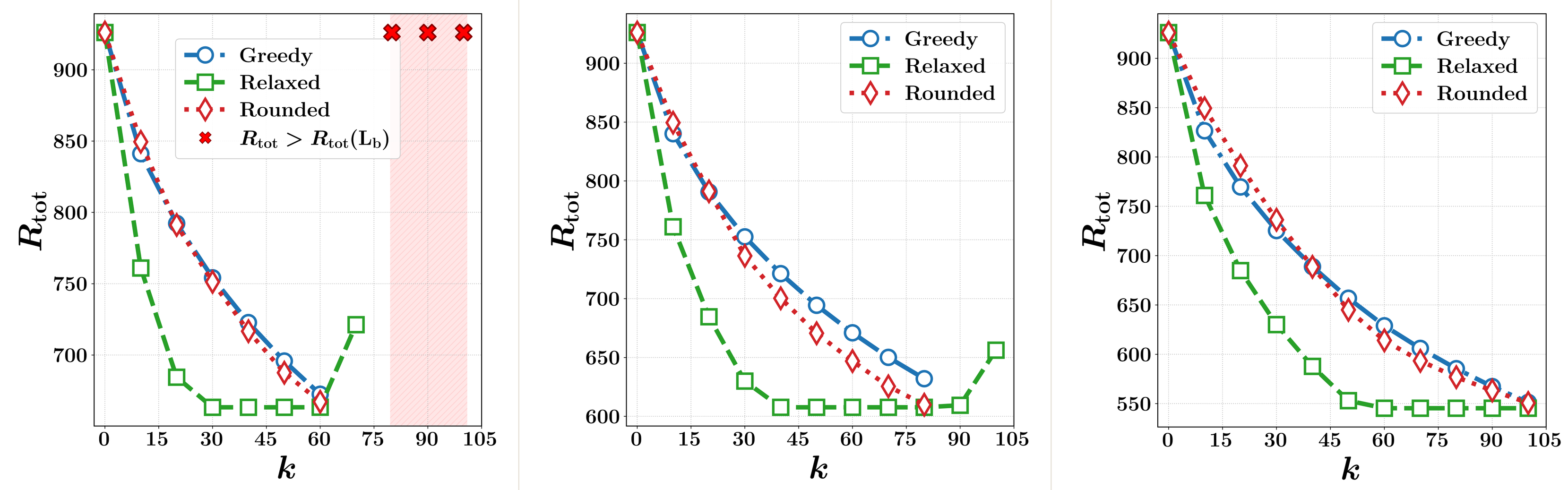

Constrained. $C = 251.69$. Infeasible from $k = 80$. Greedy stalls after 61 links.

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Relaxed. $C = 560.84$. All k feasible. Rounded tracks the bound.

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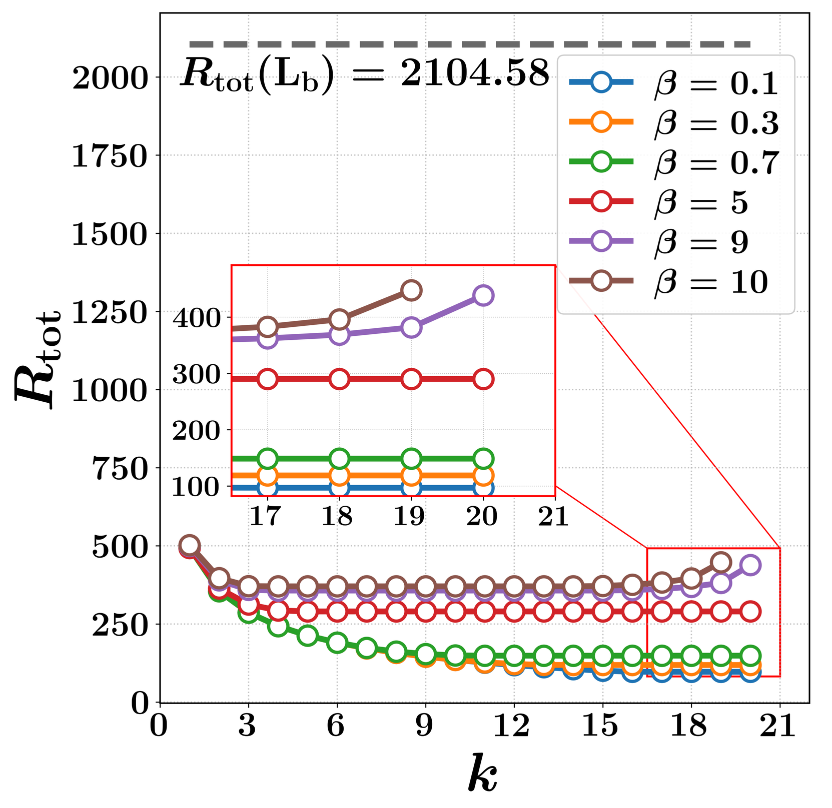

2a. $n = 500$ sampled nodes at $C = 150$, sweeping β. A steeper β lifts the curve.

2104.58

base $R_{\mathrm{tot}}(\mathbf{L_b})$ of the sampled subgraph

20 edges

for $\beta = 10$, adding 20 edges renders $(\mathcal{P}_1)$ infeasible

10 Takeaways

- The exact SDP optimum lower-bounds every feasible discrete augmentation.
- The Pareto front exposes the trade-off between effective resistance and budget spend.
- The relaxed-to-rounded ratio is an a posteriori gap certificate for each design.
- The up-to-k greedy trades solution quality for computational speed.

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Paper & code
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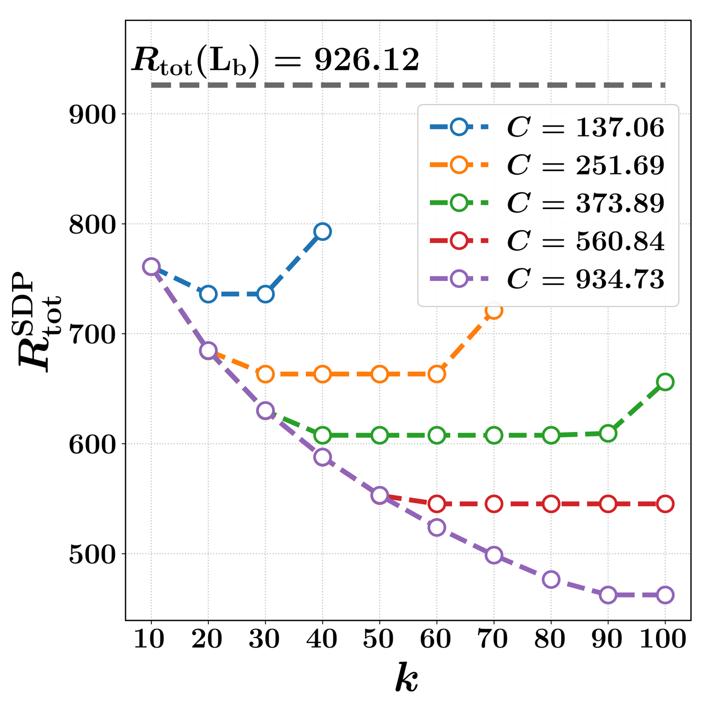

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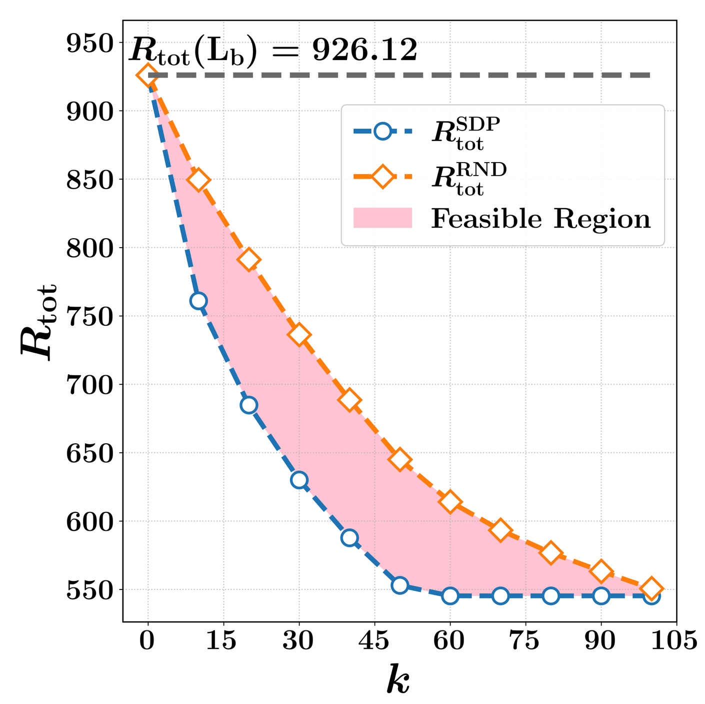

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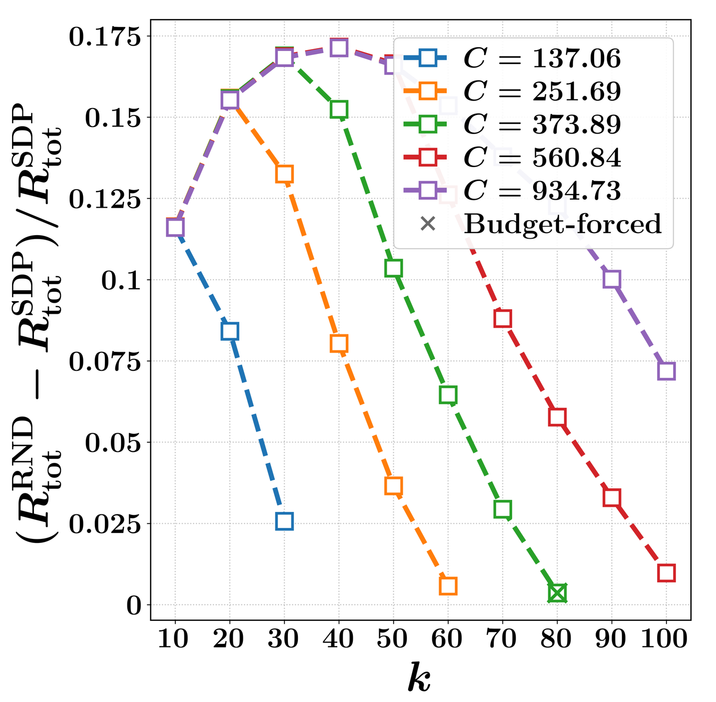

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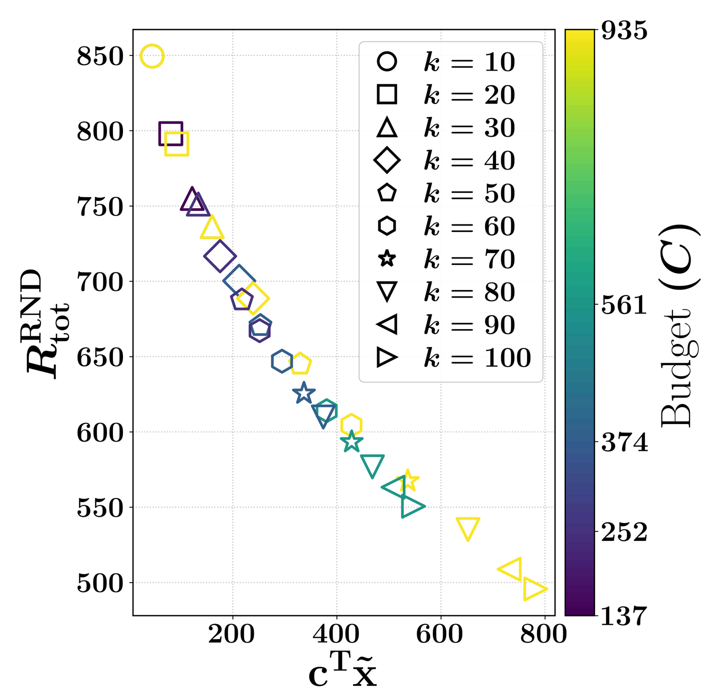

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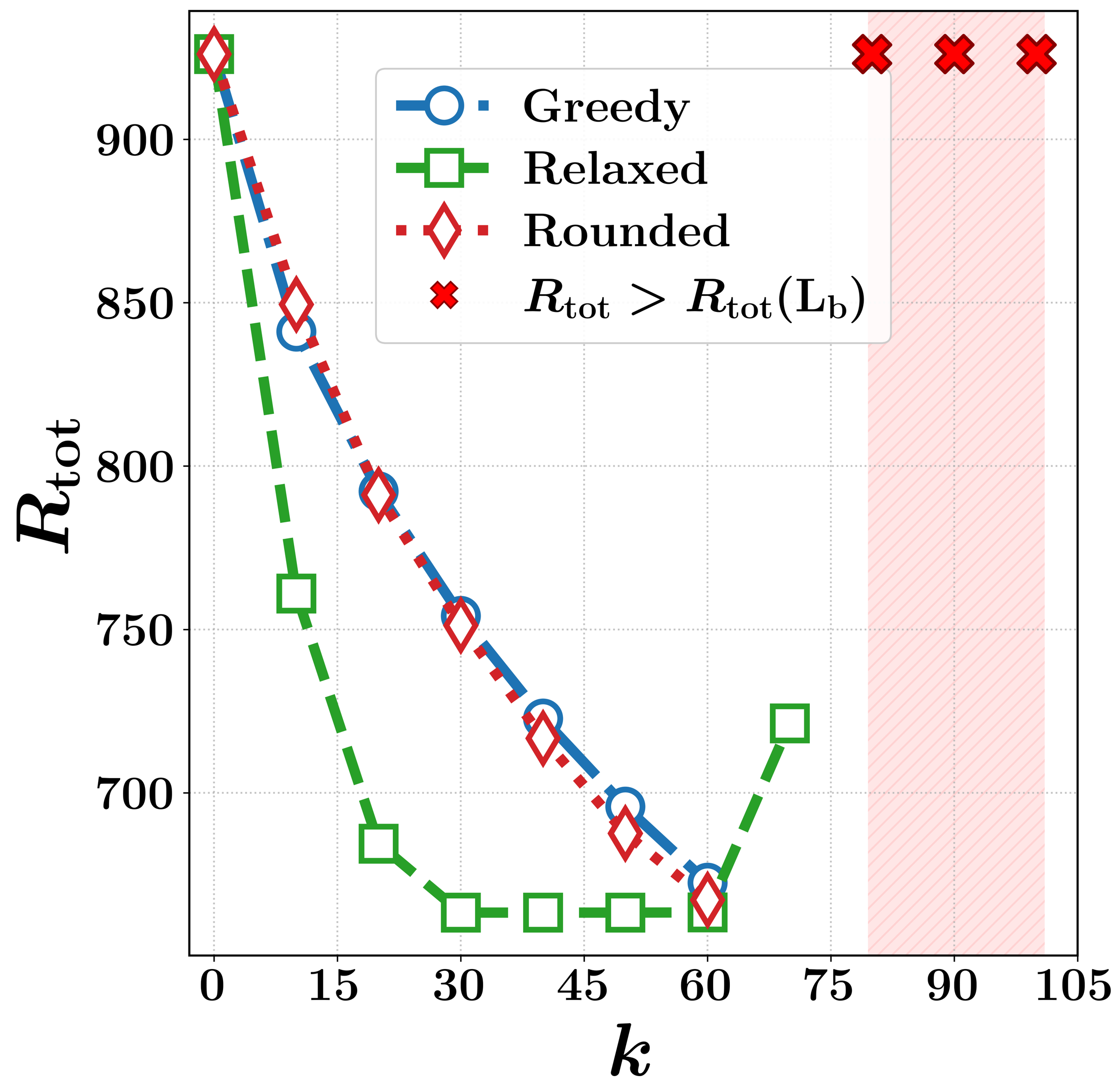

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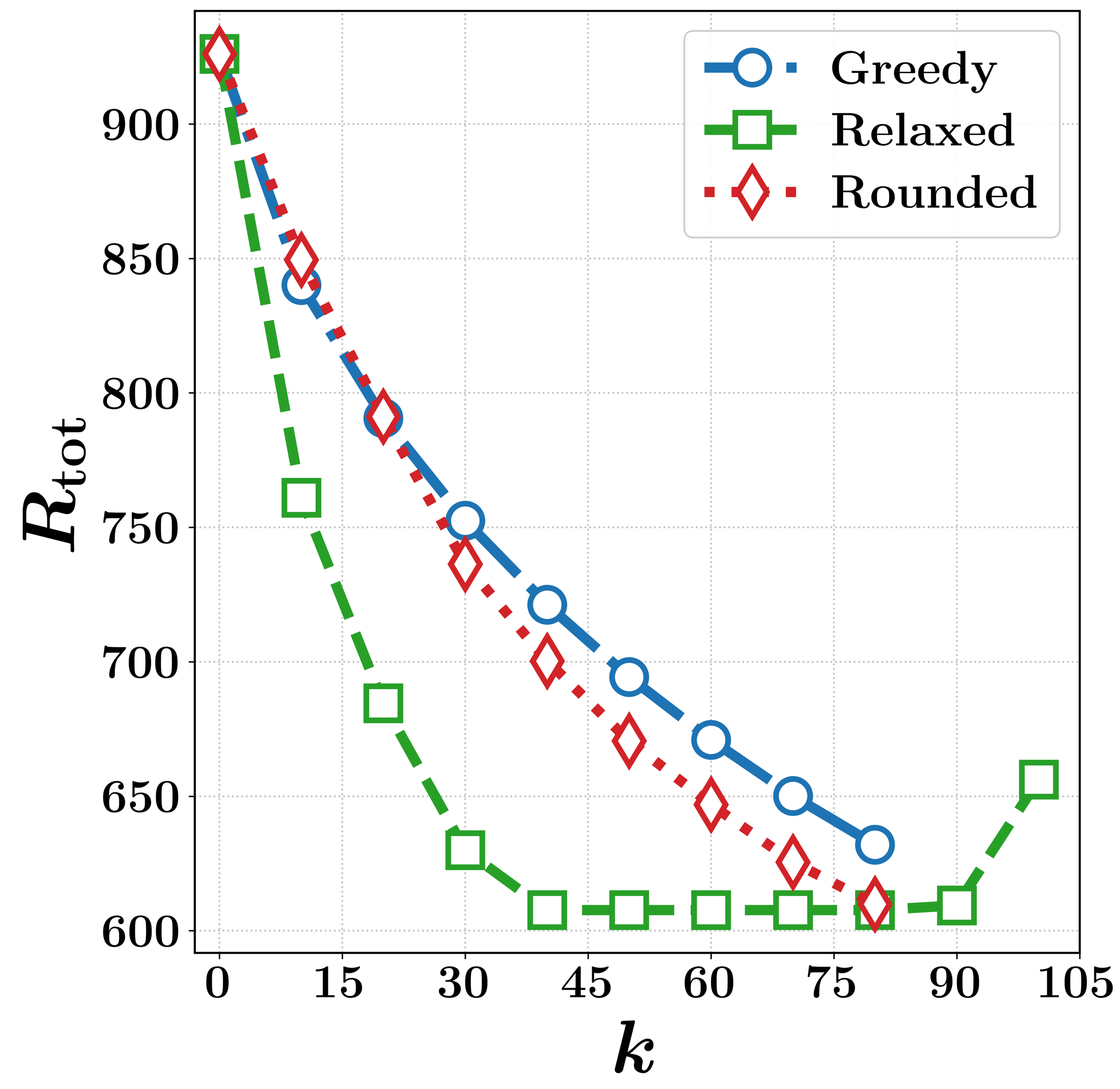

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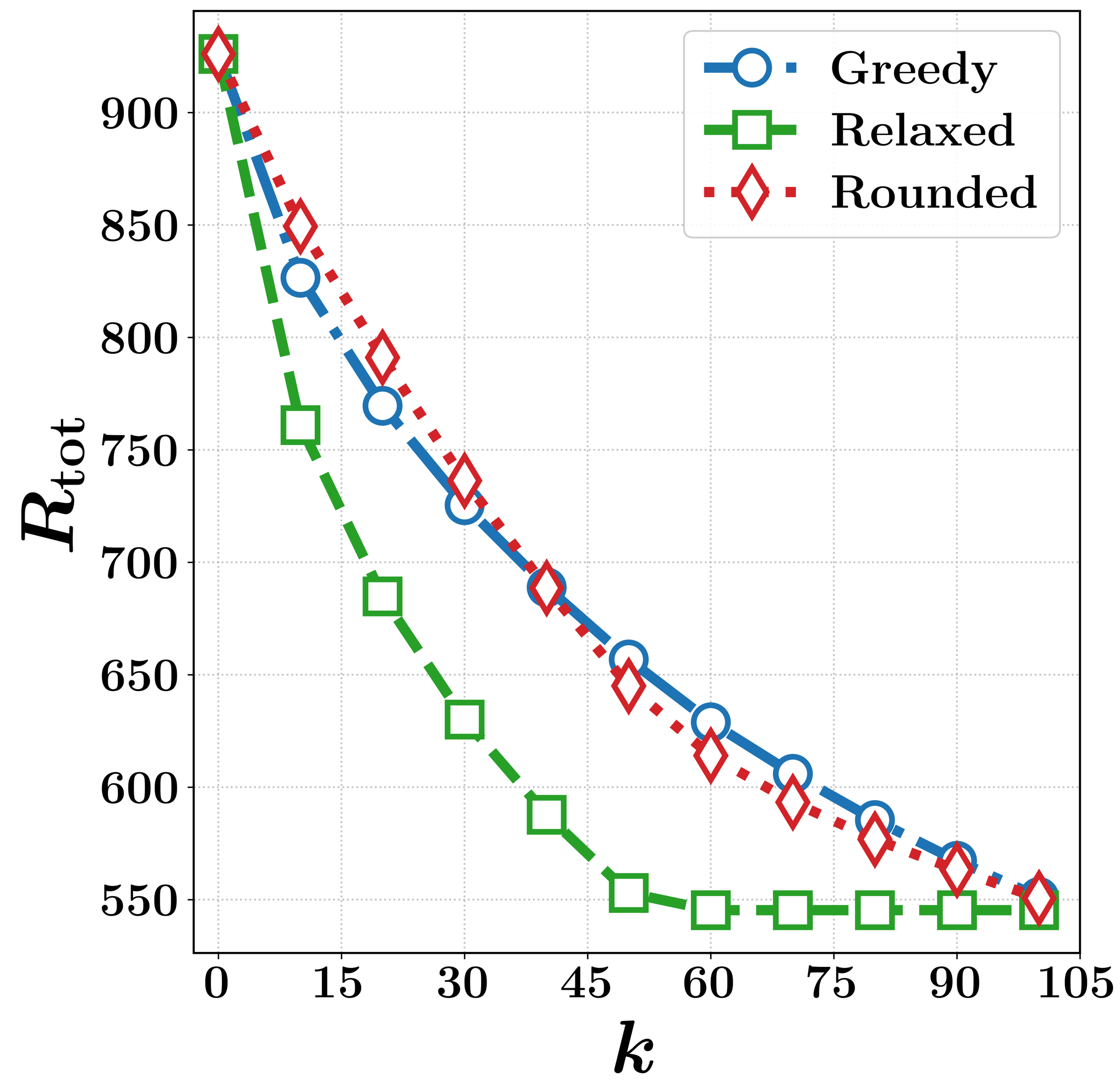

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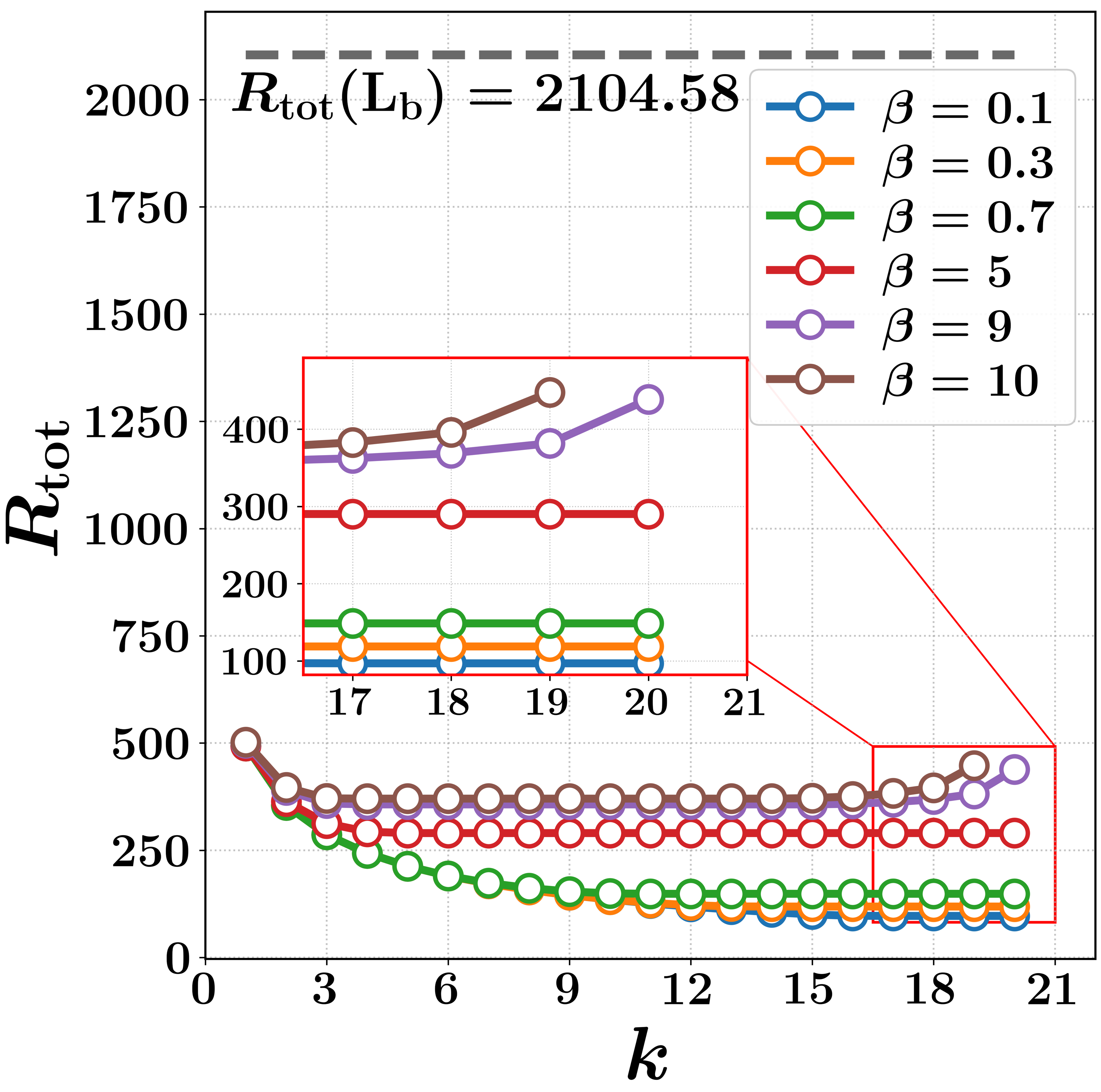

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